

Problem 10.25

A deposit is made on January 1, 2004. The investment earns interest at a constant force of interest of 6%. Calculate the monthly effective interest rate for the month of December 2004.

Solution

The force of interest is constant at

$$\delta = 0.06.$$

Therefore, the accumulation factor over a period of length t years is

$$e^{\delta t}.$$

One month represents $\frac{1}{12}$ of a year, so the monthly effective interest rate i_m satisfies

$$1 + i_m = e^{0.06/12}.$$

Hence,

$$\begin{aligned} i_m &= e^{0.06/12} - 1 \\ &= e^{0.005} - 1 \\ &\approx 0.005012520859. \end{aligned}$$

Since the force is constant, this rate is the same for every month of the year, including December. Therefore, the monthly effective interest rate is

$$\boxed{i_m \approx 0.5012521\%}.$$